

BIKE DEKHO DATA ANALYSIS PROJECT



Data Analyst Intern – Simran Sharma

TITLE: BIKE DEKHO (*CUSTOMER ANALYSIS PROJECT*)

INTERNSHIP PROJECT AT LABMENTIX

PRESENTED BY: SIMRAN SHARMA



Data Analyst Intern – Simran Sharma

AGENDA

- ▀ *Project Goal*
- ▀ *Data Cleaning and Preprocessing Process*
- ▀ *Dashboard and Charts Elaboration*
- ▀ *Final Customer Profile*
- ▀ *Business Recommendations*
- ▀ *Acknowledgement*





PROJECT GOAL:

- ✓ UNDERSTAND CUSTOMER BEHAVIOR
- ✓ IDENTIFY WHAT KIND OF CUSTOMERS ARE MOST LIKELY TO BUY A BIKE
- ✓ SUPPORT BUSINESS DECISIONS USING EXCEL-BASED ANALYSIS



Item	Description
Data Format	Excel (XLSX)
Initial Rows	1,026
Final Valid Records	1,000
Extended Attributes	15 columns
Type of Data	Structured behavioral data (cross-sectional)



Each variable / attribute below helped me understand different customer behaviors and buying patterns.

Attribute	Type	Purpose
No. of Cars	Numeric (Discrete)	Transport preference
Miles Distance	Numeric (Continuous)	Commute behavior
Bike Bought	Binary (Yes/No)	Target variable
Education	Ordinal	Awareness level
Occupation	Nominal	Economic position
Income Level	Ordinal	Spending power

DATA CLEANING PROCESS

- ✓ REMOVED DUPLICATES
- ✓ HANDLED MISSING VALUES
- ✓ STANDARDIZED CATEGORIES
- ✓ CONVERTED TO EXCEL TABLE
- ✓ PERFORMED EDA USING PIVOT TABLES & PIVOT CHARTS



GENDER ANALYSIS

Gender	Bike Buyers	% of Total Buyers
Female	239	23.90%
Male	242	24.2%

→ **Insight:**

- Gender does **not** significantly impact bike purchasing
- Both males and females purchased bikes at nearly equal rates.

This helps avoid gender-based assumptions and enables **neutral marketing strategies**.

EDUCATION ANALYSIS

Chart Type: Bar Gradient colors

→ **Insight:**

- Bachelor's degree holders bought the most bikes (35.1%)
- Followed by Partial college students
- Higher education correlates with higher purchase behavior

Education positively influences bike purchasing — possibly due to awareness, lifestyle preferences, and financial planning.

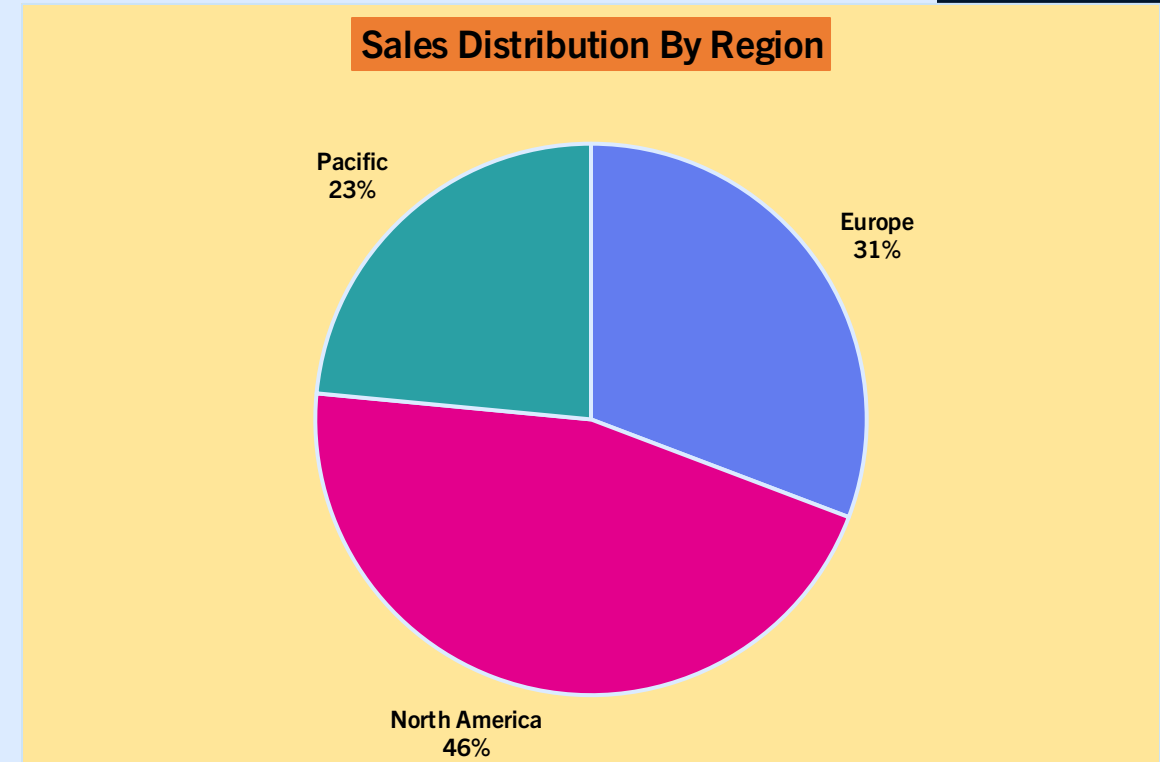
REGION-WISE SALES

CHART TYPE: PIE CHART

Region	% Sales
North America	46% (Highest)
Europe	31%
Pacific	23%

→ Insight:

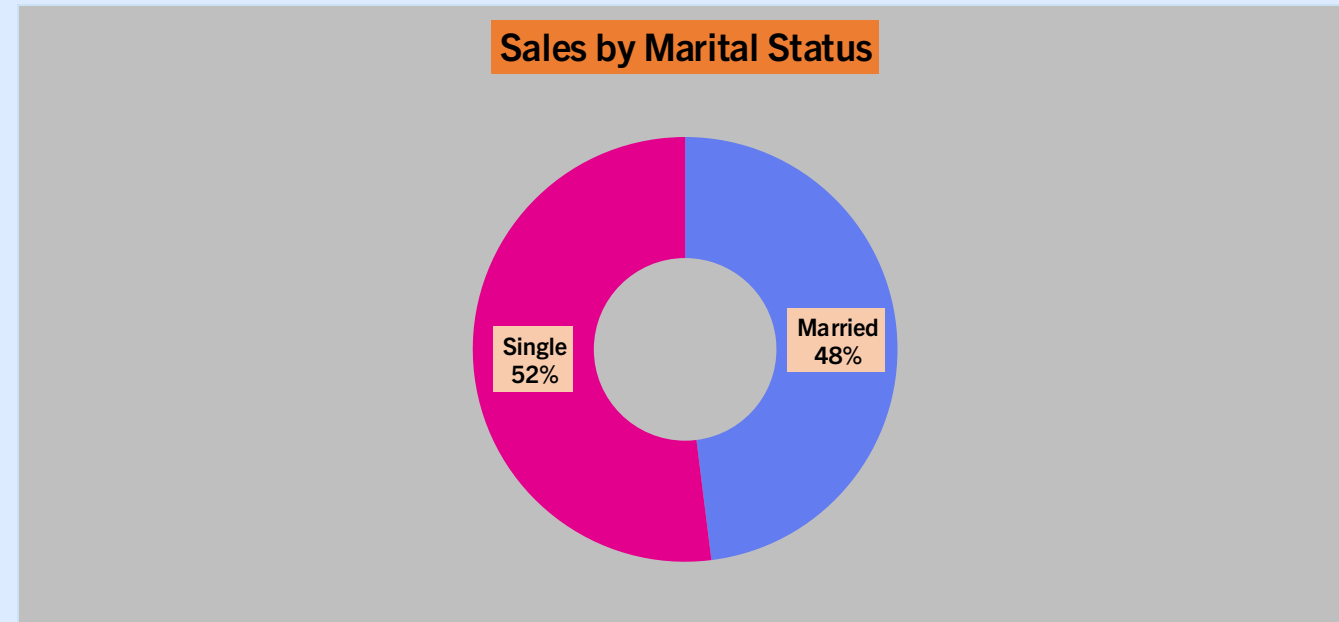
- **North America is the strongest target region** for marketing and sales efforts.
- This suggests potential for **regional campaigns and location-based offers**.



SALES BY MARITAL STATUS

CHART TYPE: DONUT CHART

Marital Status	% Buyers
Single	52%
Married	48%



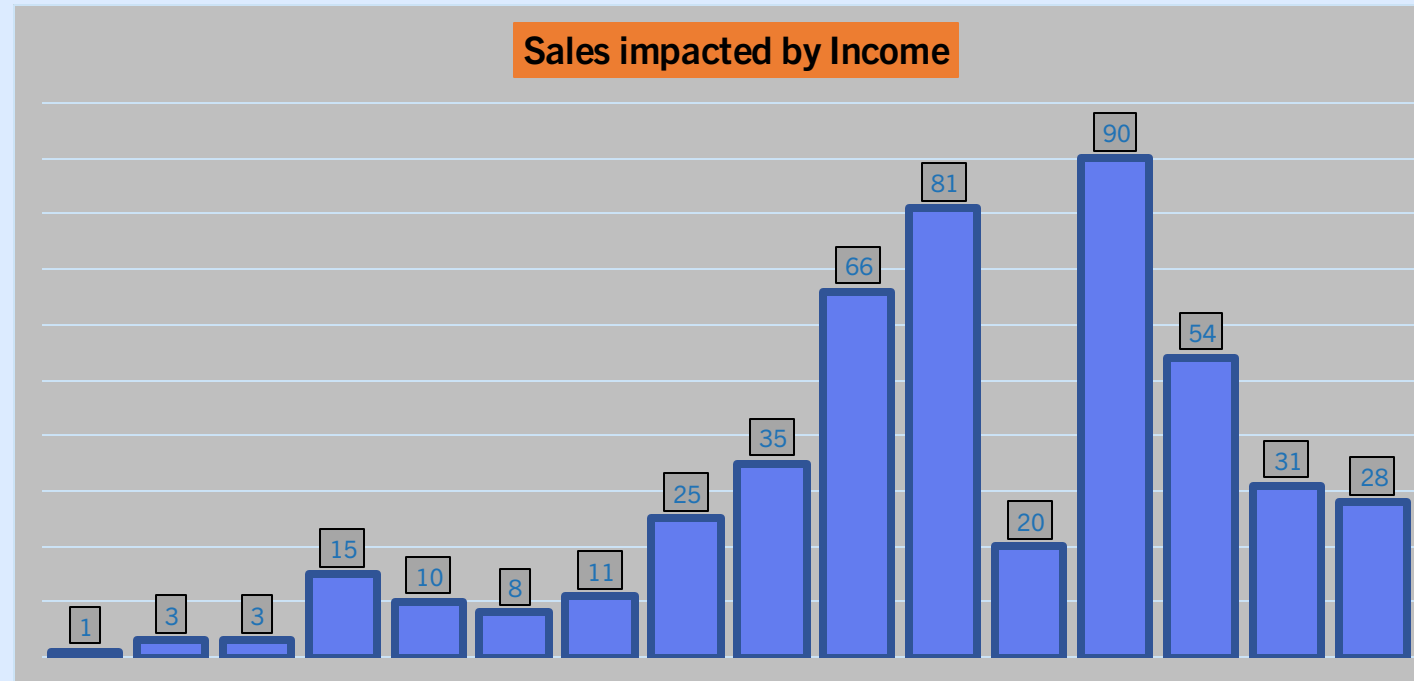
→ Insight:

- **North America is the strongest target region** for marketing and sales efforts.
- This suggests potential for **regional campaigns and location-based offers**.

INCOME IMPACT ON SALE

CHART TYPE: CLUSTERED COLUMN CHART

Income Group	% Buyers
High Income	10.6%
Medium Income	65.9%
Low Income	23.5%



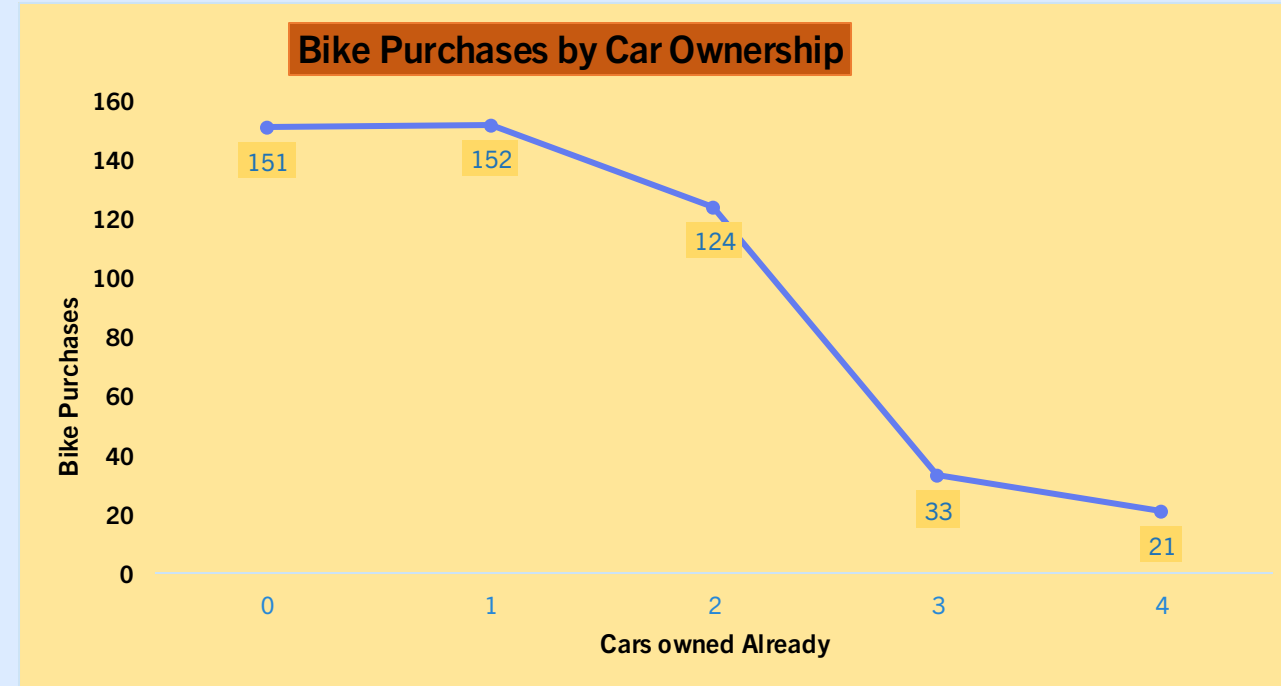
→ Insight:

- **The medium-income group is the strongest buyer segment**, showing that **cost efficiency and practicality** drive their purchasing decisions.

CAR OWNERSHIP IMPACT

CHART TYPE: PIE CHART

Cars	Purchased Bike
0	151
1	152
2	124
3	33
4	21
Grand Total	481



→ Insight:

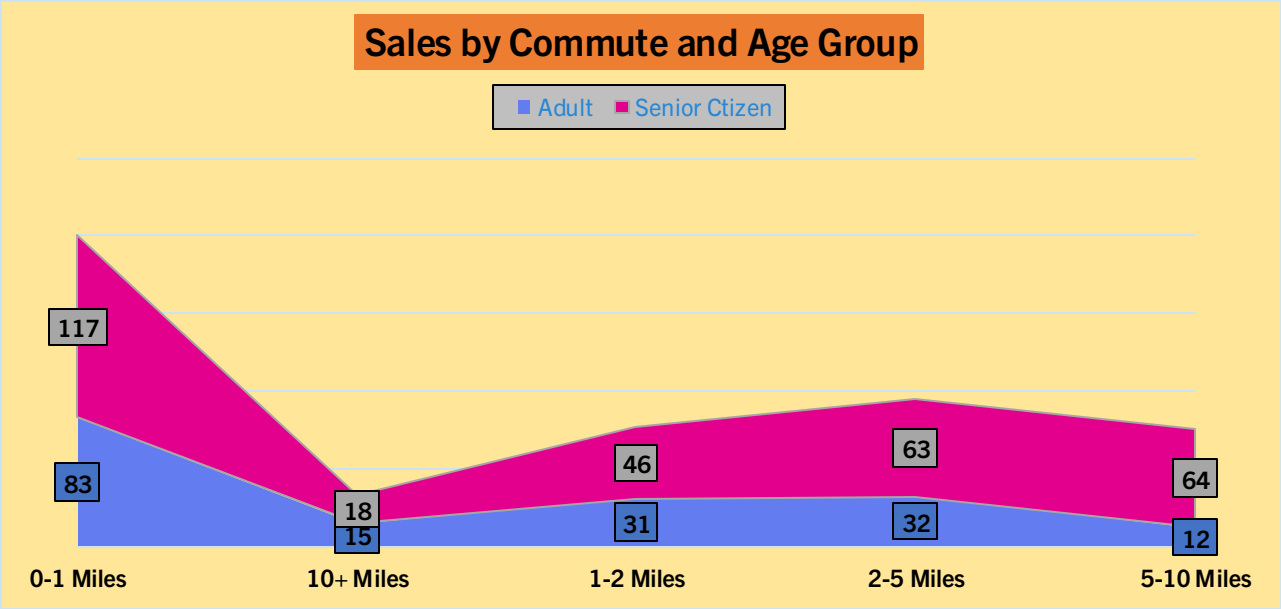
- As the number of cars increases, bike purchases decrease.
- This indicates that customers with no cars rely more on bikes as their primary mode of transport.

COMMUTE DISTANCE ANALYSIS



CHART TYPE: CLUSTERED LINE CHART

Distance (Miles)	Adults	Seniors
0–1 Miles	83	117
1–2 Miles	31	46
2–5 Miles	32	63

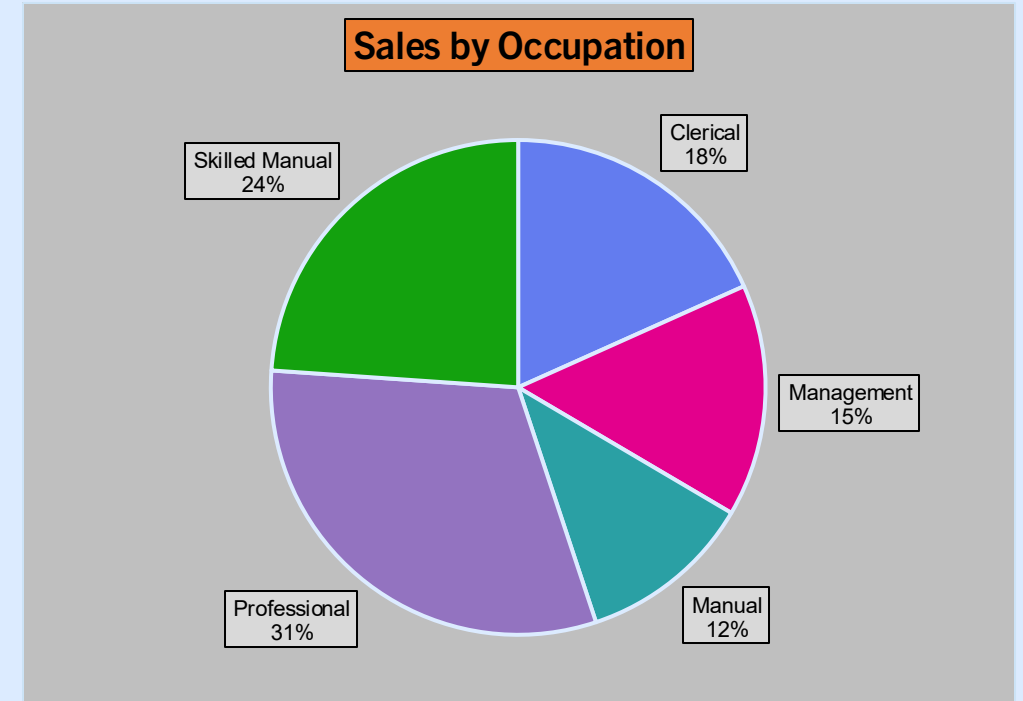


- Insight:
- Short-distance commuters (under 1 mile) are the most likely to purchase a bike
 - This suggests bikes are preferred for convenience and low-cost commuting.

IMPACT OF OCCUPATION

CHART TYPE: PIE CHART

Occupation	Bike Buyers
Professional	150
Skilled Manual	115
Clerical	88
Management	73
Manual	55
Total	481



→ Insight:

- As we can see, **professional workers lead with 150 purchases**, making them the **strongest customer segment**
- **Skilled Manual workers follow closely with 115 purchases**, which suggests that this group considers bikes a practical transport solution

THE IDEAL TARGET SEGMENTS BASED ON OCCUPATION ARE **PROFESSIONAL & SKILLED MANUAL WORKERS**, WHO TOGETHER CONTRIBUTE MORE THAN **HALF OF TOTAL BIKE SALES.**"

Female

23.90%

Male

24.2%

Total Customers 1000

% Bought Bike ↓

481 (48.1%)

Gender

Bachelors

Graduate

High School

Partial College

Partial High School

Grand Total

Female

16.6%

11.0%

7.1%

12.9%

2.1%

49.7%

Male

18.5%

8.5%

9.4%

11.9%

2.1%

50.3%

Grand Total

35.1%

19.5%

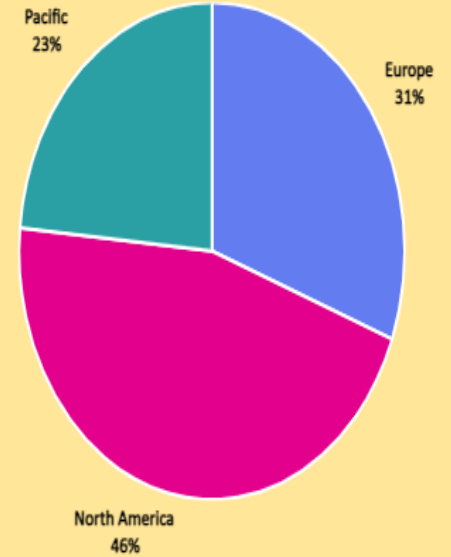
16.4%

24.7%

4.2%

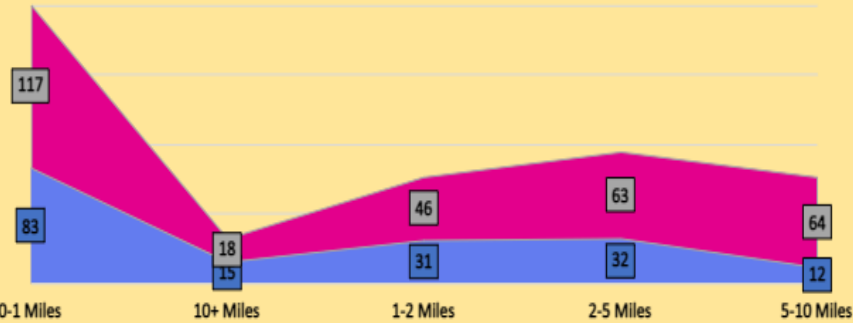
100.0%

Sales Distribution By Region

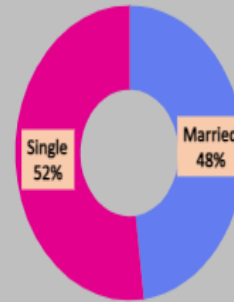


Sales by Commute and Age Group

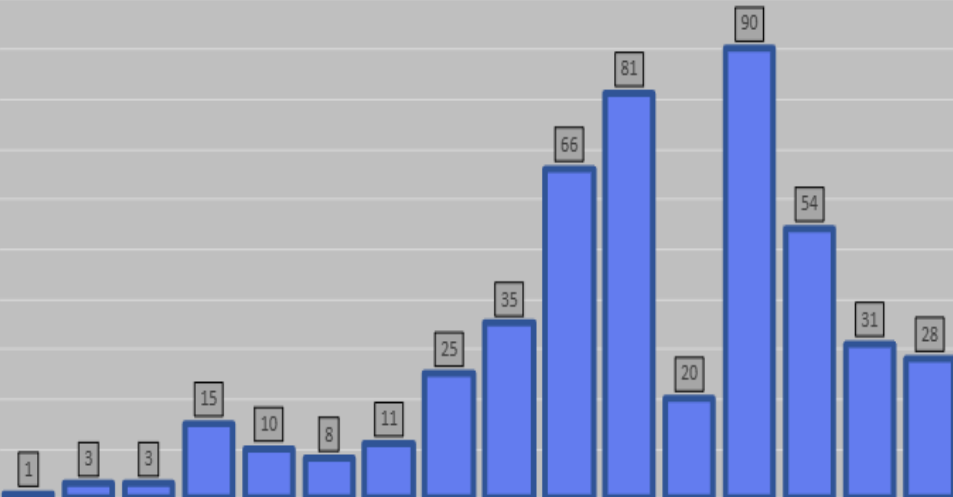
Adult Senior Citizen



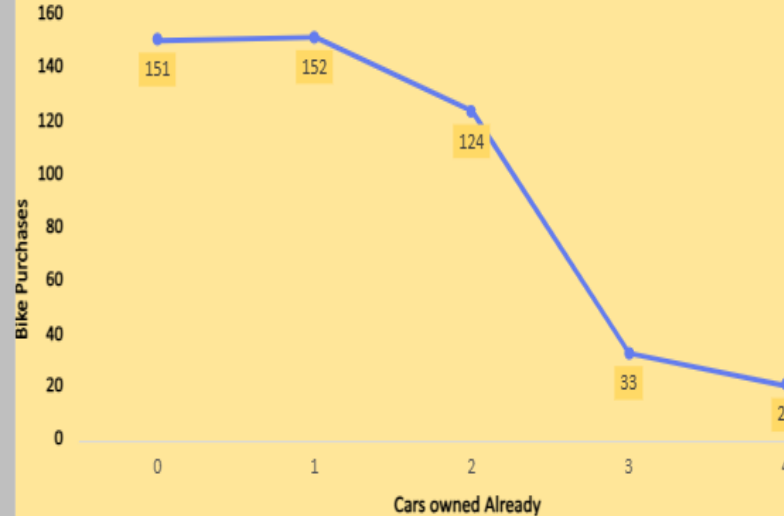
Sales by Marital Status



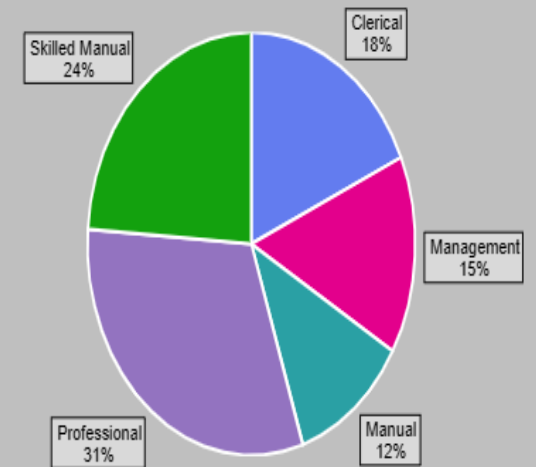
Sales impacted by Income



Bike Purchases by Car Ownership



Sales by Occupation

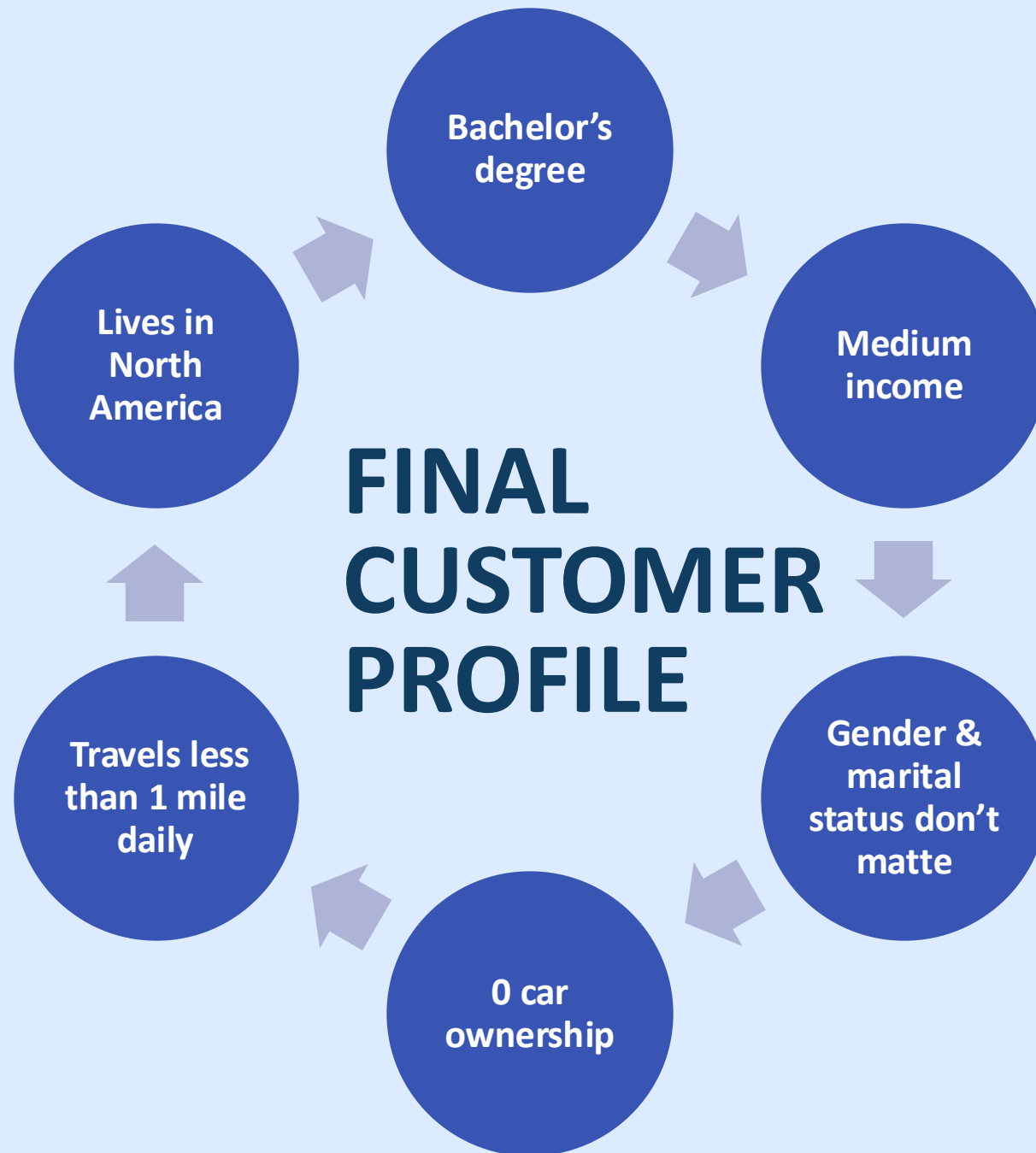


KPI DASHBOARD

KPI	Value
Total Users	1000
Total Bikes Sold	481 (48.1%)
Avg. Income	56,360
Bike vs Car Preference	Bike is 64% as popular as cars

→ Insight:

- This means **almost 1 out of every 2 customers purchased a bike**, indicating strong market potential.



LET'S TAKE YOUR BUSINESS TO THE NEXT LEVEL—TOGETHER.



Thank you!!

Thank you for giving me the opportunity to work on this project.

This analysis helped me understand real customer behavior, build a complete EDA workflow, and generate data-driven insights for business strategy.

I would be happy to answer any questions.